Enrolment No:.......................   Name of Student...........................

BENNETT UNIVERSITY
THE TIMES GROUP

NAAC GRADE A+
ACCREDITED UNIVERSITY

End Semester Examinations, May 2026

Even Semester 2025-26

B.Tech. (CSE)/ B.Tech. (CSE- Global) – 2nd Semester

2025CSET151- INTRODUCTION TO ELECTRICAL & ELECTRONICS ENGINEERING

Time: 2 hrs.   Max Marks: 40

Note: ***Student must attempt 5 Questions in total.*** *First question is compulsory, and student must attempt questions from each section. Marks are indicated against each question. Use of non-programmable scientific calculator is allowed.*

PART-A

Attempt any 6 questions from out of eight. ***2*6=12 Marks***

Q.1 (a) State the maximum power transfer theorem of electrical networks. [CO-1] [L-1] **2**

(b) Calculate the equivalent resistance between the terminals *a-b* for the electrical circuit shown in Fig. 1. [CO-1] [L-3] **2**

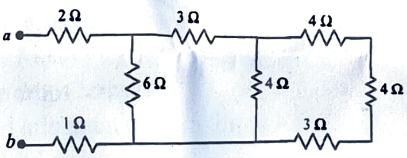

Fig. 1

(c) Explain the input-output relationship of the R-L high pass filter. [CO-2] [L-1] **2**

(d) Explain the operation of the Zener Diode as a voltage regulator using the appropriate circuit diagram. [CO-3] [L-1] **2**

(e) Write the P-N junction diode current equation and explain the variables. [CO-3] [L-2] **2**

(f) At what reverse bias voltage does the reverse bias current in a PN Junction reaches 90% of its reverse saturation current? [CO-3] [L-3] **2**

(g) Define common mode rejection ratio (CMRR) of the Op-Amp and what is CMRR value of an ideal Op-amp. [CO-4] [L-2] **2**

(h) Explain virtual short (virtual ground) concept of the Operational Amplifier along with the circuit diagram. [CO-4] [L-] **2**

PART-B

Attempt any 2 questions from this part B.

Q.2 Using nodal analysis, find V_0 in the circuit of Fig. 2. [CO-1] [L-4] 6

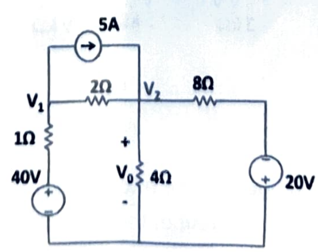

Fig. 2

Q.3 For the diode circuit shown in Fig. 3, $V_S = 2$ V. The silicon diode has a reverse saturation current of 1 nA at 300 K. Given that V = 0.7 V. Assume ideality factor=1.

a. Find R_2 when $R_1 = 1$ kΩ
b. Find R_1 when $R_2 = 1$ kΩ

[CO-3] [L-3] 6

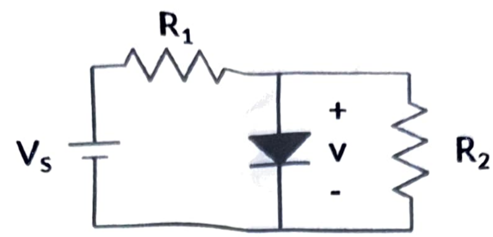

Fig. 3

Q.4 Consider a summing amplifier shown in Fig. 4. Given that $R_1 = 1$ kΩ, $R_2 = 1.5$ kΩ, $R_3 = 2$ kΩ, and $R_F = 10$ kΩ. If $V_{I1} = 0.5$ V, $V_{I2} = 0.75\sin(60\pi t)$ and $V_{I3} = 1.0\sin(30\pi t)$, find the output voltage? [CO-4] [L-3] 6

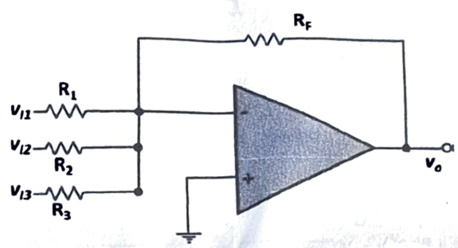

Fig. 4

*6*2=12 Marks*

PART-C

Q.5 For the circuit shown in Fig. 5, identify the filter type, find the transfer function and cut-off frequency of the filter. The output of the filter is taken between nodes C and D. [CO-2] [L-3] **8**

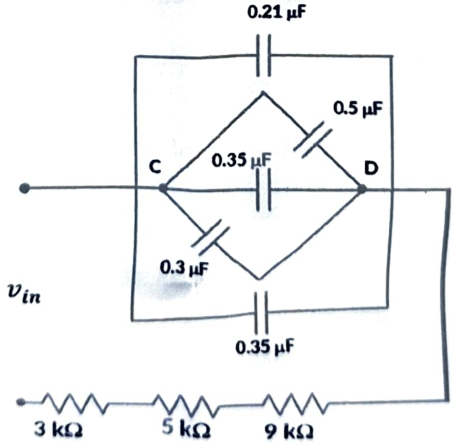

Fig. 5

OR

For the circuit shown in Fig. 6, identify the filter type, find the transfer function and cut-off frequency of the filter. The output of the filter is taken between nodes C and D. [CO-2] [L-3] **8**

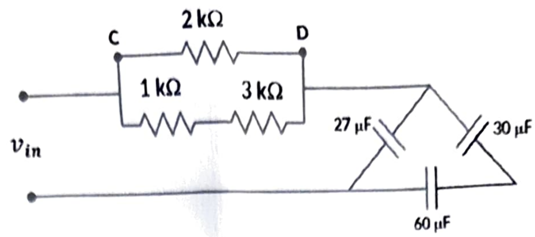

Fig. 6

Q.6 For the diode circuit (Si diodes with cut in voltage of 0.6 V) shown in Fig. 7, let R_f = 1 kΩ and R = 9 kΩ. Find the output voltage V_0 when

i) $V_1 = V_2 = 0$ V
ii) $V_1 = 10$ V, $V_2 = 0$ V
iii) $V_1 = V_2 = 10$ V

[CO-3] [L-4] **8**

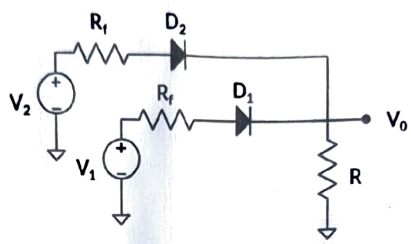

Fig. 7

OR

Consider the circuit shown in Fig. 1. Given that, $R = 4$ kΩ, $R_L = 10$ kΩ and V_Z =30 V. The input voltage V_{in} can vary between 70 to 100 V. Answer the following questions.
i) What is the range of voltage drops across R?
ii) Calculate the maximum and minimum currents through R?
iii) What is the current in the load resistor?
iv) Calculate the maximum and minimum currents through the Zener diode.
v) What is the minimum power rating of Zener diode that needs to be used in the circuit?

[CO-3] [L-4] **8**

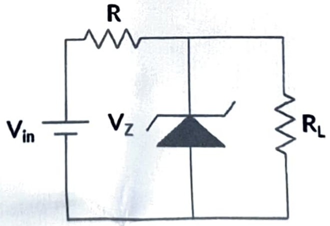

Fig. 8

8*2=16 Marks

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